

Run 01 | TEA

Global (including India)

IV only

Target Total early-stage entrepreneurial activity; binary 1 vs. 0

Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022

Positive cases 154,420 (11.50%) **Case time** 22m 22s

Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)

Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9881). Highest F1: XGBoost (0.8488). The all-negative accuracy baseline is 88.50% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9881	0.8906	96.29	79.81	90.65	0.8488	15s
CatBoost	0.9881	0.8903	96.28	79.77	90.63	0.8486	1m 16s
Random Forest	0.9880	0.8889	96.29	79.84	90.59	0.8488	1m 28s
H2O GBM	0.9876	0.8849	96.25	79.79	90.22	0.8469	26s
Elastic Net	0.9873	0.8827	96.16	79.39	89.96	0.8435	5m 49s
Lasso	0.9873	0.8826	96.17	79.42	89.97	0.8437	5m 01s
Logistic Regression	0.9873	0.8828	96.17	79.43	89.99	0.8438	46s
Ridge	0.9872	0.8821	95.98	76.65	93.52	0.8425	4m 56s
Decision Tree	0.9835	0.8136	96.20	79.55	90.08	0.8449	5s
Naive Bayes	0.9360	0.5577	90.52	55.56	87.51	0.6797	1m 55s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	82,481.50	63.03
2	Established business owner	29,131.64	22.26
3	Self-efficacy	4,737.20	3.62
4	Knowing entrepreneurs	3,767.75	2.88
5	Self-efficacy (missing indicator)	1,607.63	1.23
6	Opportunity perception	1,472.16	1.13
7	Business exit	1,022.00	0.78
8	Business angel	771.03	0.59

Share is the percentage of total forest impurity importance across all 25 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Source: Completed Objective 1 batch, 27 September 2026; 01_obj1_global_iv_TEA. Scores and importance are taken directly from saved CSV outputs.